

Template Week 5 – Operating Systems

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Assignment 5.1: Unix-like

- a) Find out what the difference is between UNIX and unix-like operating systems?

UNIX is the *original* operating system created in the 1970s at Bell Labs. It is a certified, commercial system.

A **Unix-like** operating system is *not the original UNIX*, but it behaves like UNIX, follows similar concepts, commands, and POSIX standards. Examples: **Linux**, **macOS**, **BSD** (shown on page 13 as “Unix-like” systems)

- b) Study the image above named UNIX timeline. Find out who Ken Thompson, Dennis Ritchie, Bill Joy, Richard Stallman, and Linus Torvalds are and what they have contributed to the development of UNIX or unix-like systems and to IT in general. **TIP!** English-language sources often contain more detailed information about these individuals.

Ken Thompson

One of the creators of the original **UNIX** at Bell Labs. He helped design early UNIX concepts and tools (shell, filesystem ideas).

Dennis Ritchie

Co-creator of **UNIX** and inventor of the **C programming language**, which allowed UNIX to be rewritten in C. This made UNIX portable and influenced almost all modern OSes.

Bill Joy

Developer of **BSD UNIX** at Berkeley. He created the **vi editor** and contributed to networking features in BSD that influenced the internet.

Richard Stallman

Founder of the **GNU movement**. He started the idea of free software and wrote important tools like the GNU Compiler Collection (GCC). His project aimed to build a completely free UNIX-like system.

Linus Torvalds

Creator of the **Linux kernel** in 1991. When combined with GNU tools, it formed **GNU/Linux**, the most widely used Unix-like OS today

- c) What is the philosophy of the GNU movement?
- The GNU movement is based on the idea that software should be free—not in price, but in freedom.
 - It promotes the user’s rights to run, study, modify, and share software.

- c. Its goal is to ensure that computing is controlled by users, not by companies or restrictive licenses.

- d) Does Ubuntu as a Linux operating system conform to the philosophy of the GNU movement? Please explain your answer.

Ubuntu supports the GNU ideas but is not completely free software because it includes some non-free components.

- e) Find out what is the Windows Subsystem for Linux?

WSL is a Windows feature that allows users to run a **Linux environment inside Windows** without using a virtual machine. It lets you run Linux commands, tools, and even graphical apps directly on Windows.

- f) Find out, which operating system family belongs to Android, iOS and ChromeOS?

Android → Unix-like (built on the Linux kernel)

iOS → Unix-like (based on Darwin/BSD)

ChromeOS → Unix-like (also built on the Linux kernel)

So all three belong to the Unix-like family.

Assignment 5.2: Supercomputers and gameconsoles

- a) Research on this site what supercomputers are used for and write a short summary of it:

<https://www.computerhistory.org/timeline/search/?q=Supercomputer>

Supercomputers are extremely powerful computers designed to do a huge number of calculations very quickly. They are used for tasks that normal computers cannot handle. According to the Computer History Museum website, supercomputers are mainly used for things like **weather prediction, scientific research, and complex simulations**. Scientists use them to model climate change, study molecules, or simulate physics problems. Supercomputers are also used in engineering, for example to test car designs by running virtual crash or stress simulations. In addition, they have been used in areas like national security and cryptography because of their high processing power. Overall, supercomputers help solve very large and complex problems that require massive computing speed.

- b) IBM is a company that has already built a number of supercomputers. One of them is IBM's Roadrunner. The CPU developed for this supercomputer was further developed at a later stage as the CPU for the PlayStation 3 console. Find out what a **PlayStation 3 cluster** is and what it was used for?

A PlayStation 3 cluster is a group of many PlayStation 3 (PS3) consoles connected together so they work like one powerful computer. The PS3 used the **Cell processor**, which was originally related to the same technology used in IBM's Roadrunner supercomputer. Because the PS3 had a strong CPU for its time, researchers discovered they could link hundreds of PS3s to create a low-cost supercomputer.

These PS3 clusters were used for scientific research, such as **physics simulations, image processing, and military research**. One famous example is the **U.S. Air Force**, which built a cluster of over 1,700 PS3 consoles to create a very powerful and cheap supercomputer. Universities also used PS3 clusters for experiments in parallel computing and for running complex calculations.

- c) You can build a supercomputer by putting a few computers together in a cluster. Here's what Oracle did with a collection of Raspberry Pi's, for example:

<https://blogs.oracle.com/developers/post/building-the-worlds-largest-raspberry-pi-cluster>

What specific operating system is running on this cluster?

The Raspberry Pi cluster built by Oracle uses the operating system **Oracle Linux**. This is a Linux-based operating system that Oracle often uses for servers and large computing projects. So even though the cluster is made from small Raspberry Pi devices, it still runs a full Linux system designed for high-performance and enterprise workloads.

- d) Does Oracle's Raspberry Pi supercomputer appear in the list of the 500 fastest supercomputers in the world? Make a logical decision for this, without going through the entire list.

<https://www.top500.org/lists/top500/list/2023/06/>

No, Oracle's Raspberry Pi supercomputer does not appear in the TOP500 list. The TOP500 list contains the fastest supercomputers in the world, and they use very powerful CPUs and GPUs. A Raspberry Pi cluster is made from many small, low-power devices, so even with many of them combined, the performance is far too low to compete with real supercomputers. Because of that, it is logical that Oracle's Raspberry Pi cluster is not included in the TOP500.

- e) What CPU architecture is used for the PlayStation 5 and Xbox Series X?
What operating systems run on these consoles?
What conclusion can you draw from the answer to the previous question?

CPU architecture:

Both the **PlayStation 5** and the **Xbox Series X** use **AMD x86-64** CPU architecture.

This is the same architecture used in most modern desktop and laptop PCs.

- **Operating systems:**

- **PlayStation 5:** Runs a customized **Orbis OS**, which is based on **FreeBSD** (a Unix-like operating system).
- **Xbox Series X:** Runs a customized **Windows-based OS**, part of the Xbox OS family, built on the **Windows NT** kernel.

- **Conclusion:**

Even though these consoles are gaming devices, they use **the same CPU architecture as regular PCs (x86-64)** and run operating systems that are closely related to **Unix-like systems (PS5)** or **Windows (Xbox)**.

This shows that modern consoles are basically **specialized computers**, not very different from normal PCs in terms of hardware and OS design.

Assignment 5.3: Working with Windows

Take relevant screenshots of the assignments below

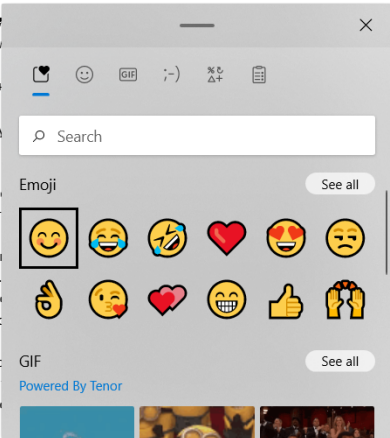
- a) Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.

Assignment 5.3: Working with Windows

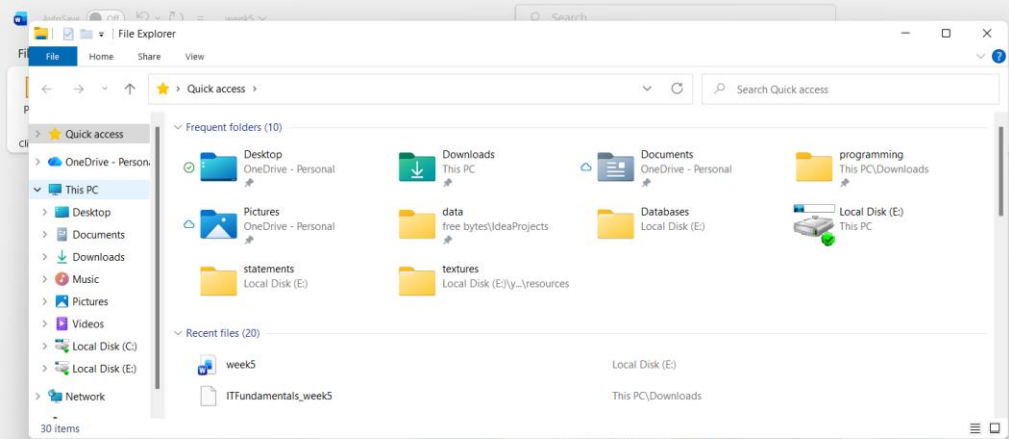
Take relevant screenshots of the assignments below

- Practice for about 10 minutes with the  keyboard shortcuts combinations, skip the general shortcuts in this exercise. Take a look at which screens are opened.
- The file explorer can be opened with  + E. Which key combination could you also use?
- Open the system properties with a  key combination. Take a screenshot of the open screen. Paste this screenshot into this template.
- Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.
- If you're giving a PowerPoint presentation, you can use the projector as a second screen. Which screen that you don't show over the projector, or the second screen. Which screen?
- If you leave the classroom for a while and your Apps will continue to run in the background. Your Apps will continue to run in the background. Which Apps will continue to run in the background? Take a screenshot of this result and paste it into this template.
- Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.

Working in the File Explorer



- b) The file explorer can be opened with  + E, Which key combination could you also use?



Working in the File Explorer

- The file explorer can be opened with  + E, Which key combination could you also use?
- Open the system properties with a  key combination, take a screenshot of the open screen. Paste this screenshot into this template.
- Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

- c) Open the system properties with a  key combination, take a screenshot of the open screen. Paste this screenshot into this template.

- Apps and Features
- Mobility Center
- Power Options
- Event Viewer
- Device Manager
- Network Connections
- Disk Management
- Computer Management
- Windows PowerShell
- Windows PowerShell (Admin)
- Task Manager
- Settings
- File Explorer
- Search
- Run
- Shut down or sign out >
- Desktop

- You forgot your power adapter, your laptop normally needs 15 watts. You will be loaned a power adapter that can deliver 50 watts. Voltage, polarity, etc. are all the same compared to the original power adapter. You can connect the borrowed power adapter to your laptop. What will happen? Also explain why you think that.

Assignment 3.4: Build your dream PC

Go to the site below and choose your own PC components to build your dream PC:

<https://nl.pcpartpicker.com/list/>

Take a screenshot of the result. Justify why you put together this configuration. Compare your current laptop/computer with the configuration of your dream PC. Describe these differences in detail.




- d) Open task manager with a key combination. Take screenshots of the tabs: processes (shows active processes), performance, and users. Place these three screenshots in this template.

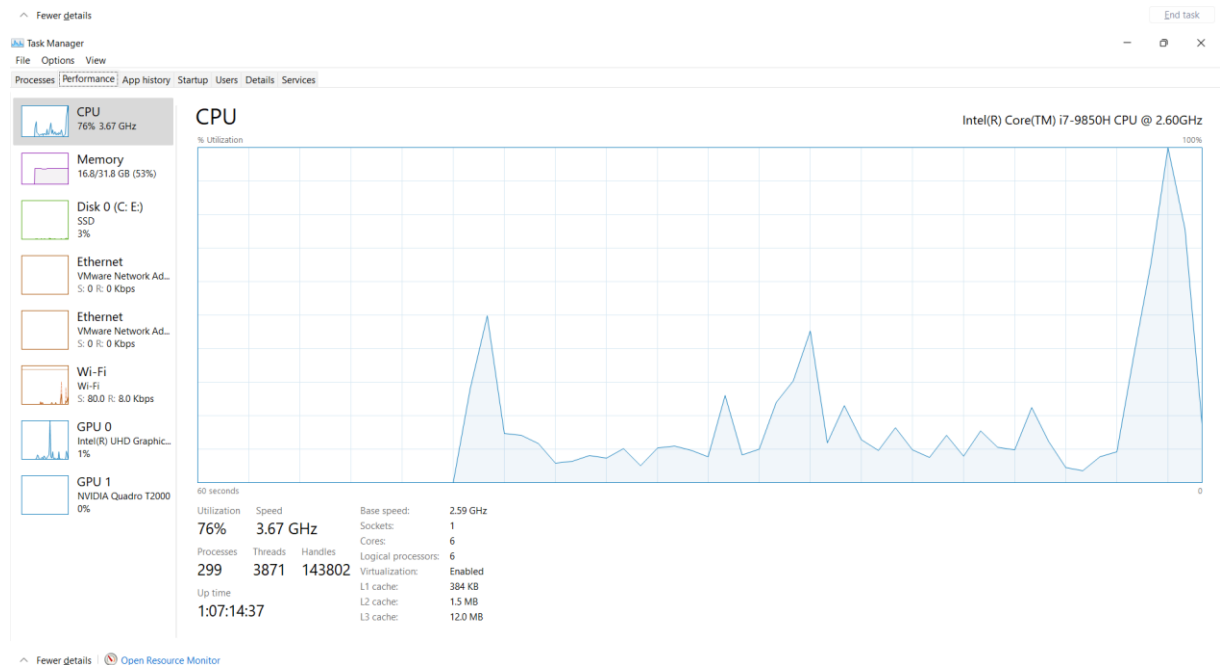
Task Manager

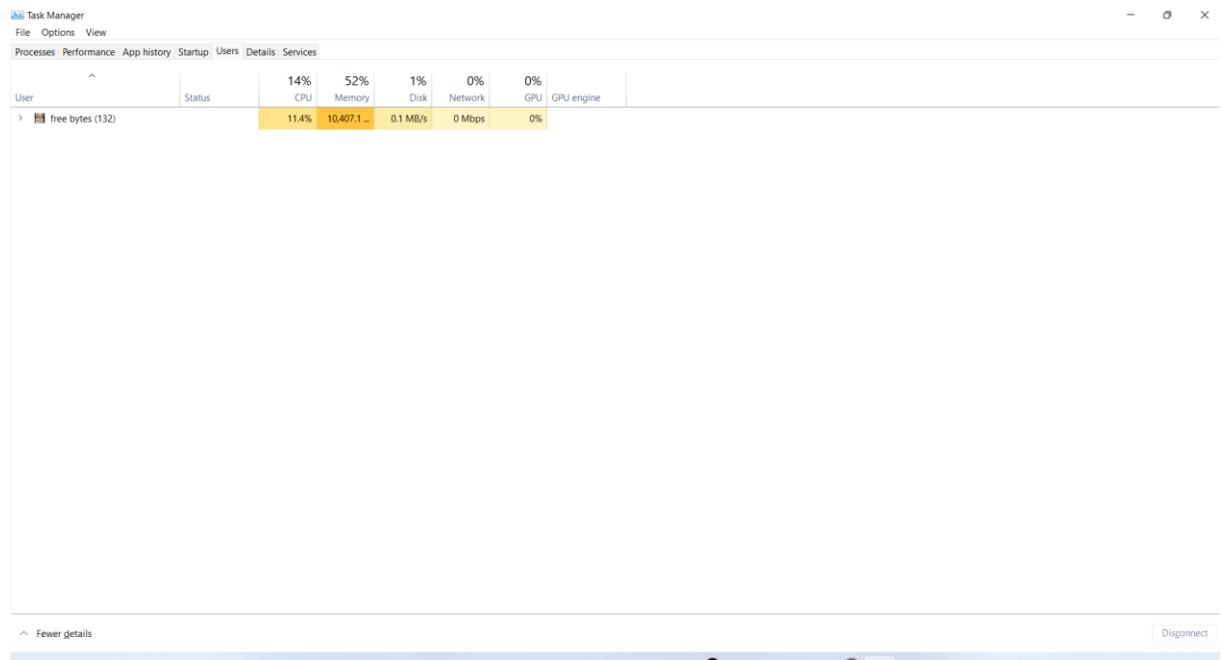
File Options View

Processes Performance App history Startup Users Details Services

Name	Status	10% CPU	51% Memory	1% Disk	0% Network	1% GPU	GPU engine	Power usage	Power usage tr...
Apps (6)									
Google Chrome (32)		1.0%	8,861.8 MB	0.1 MB/s	0.1 Mbps	0%		Very low	Very low
Microsoft Word (2)		1.0%	222.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
OpenDK Platform binary		0%	51.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
pgAdmin 4 (2)		0.1%	12.9 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Task Manager		2.3%	32.9 MB	0 MB/s	0 Mbps	0%		Low	Very low
Windows Explorer		0%	47.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Background processes (133)									
AggregatorHost		0%	0.5 MB	0 MB/s	0 Mbps	0%		Very low	Very low
AppHelperCap		0%	0.9 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Application Frame Host		0%	6.1 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus		0%	0.4 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus		0%	2.3 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus		0.2%	16.7 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus		0%	3.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus		0%	12.6 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Avast Antivirus engine server		0.1%	58.0 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Avast remediation exe		0%	0.2 MB	0 MB/s	0 Mbps	0%		Very low	Very low
Avast Service		0.4%	40.6 MB	0.1 MB/s	0 Mbps	0%		Very low	Very low
Avast Software Analyzer		0.1%	31.0 MB	0 MB/s	0 Mbps	0%		Very low	Very low
COM Surrogate		0%	1.8 MB	0 MB/s	0 Mbps	0%		Very low	Very low
COM Surrogate		0%	2.6 MB	0 MB/s	0 Mbps	0%		Very low	Very low

End task





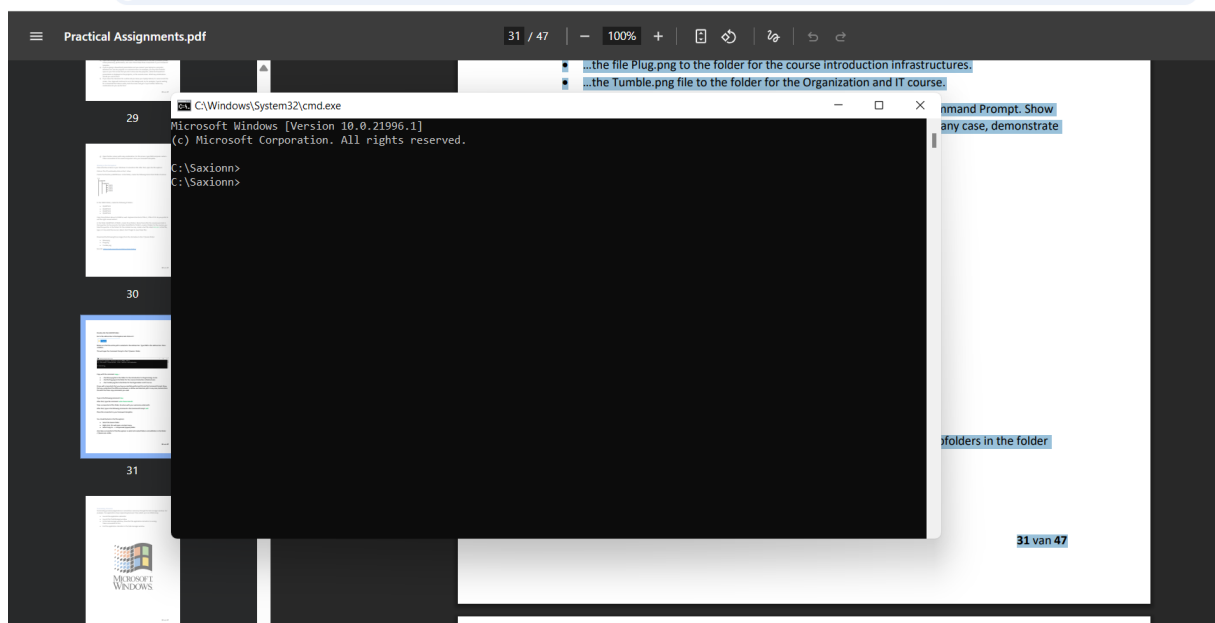
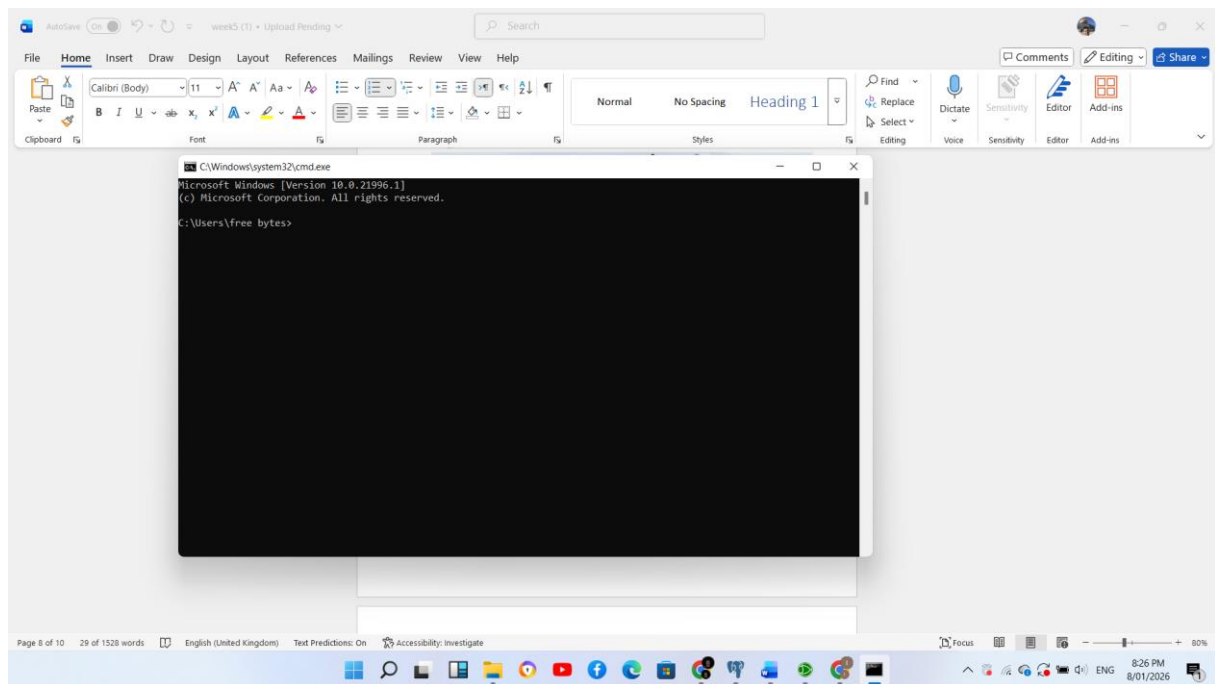
- e) If you're giving a PowerPoint presentation and you connect your laptop to a projector, Windows can use the projector as a second screen. For example, you may have Outlook open on your first screen that you don't show over the projector, while the PowerPoint presentation is displayed on the projector, or the second screen. Which key combination should you use for this?

Windows Key + P

- f) If you leave the classroom for a while and you leave your laptop behind, it is wise to lock the screen. Your Apps will continue to run in the background. So, for example, if you're waiting for a download that takes a while, lock the screen and get a cup of coffee. Which key combination do you use for this?

Windows Key + L

- g) Open the Run screen with a key combination. On this screen, type CMD and press <enter>. Take a screenshot of this result and paste it into this template.



Working in the File Explorer

Relevant screenshots **copy** command:

```
C:\Windows\System32\cmd.exe
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy C:\Saxionn\Wave.png C:\Saxionn\YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>Wave.png C:\Saxionn\YEAR1\QUARTILE1\IntroductionToProgramming
'Wave.png' is not recognized as an internal or external command,
operable program or batch file.

C:\Saxionn>
C:\Saxionn>Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
C:\Saxionn>Plug.png YEAR1\QUARTILE1\IntroductionInfrastructure
C:\Saxionn>copy Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy Wave.png HBO-ICT\YEAR1\QUARTILE1\IntroductionToProgramming
1 file(s) copied.

C:\Saxionn>
C:\Saxionn>copy Plug.png HBO-ICT\YEAR1\QUARTILE1\IntroductionInfrastructure
1 file(s) copied.

C:\Saxionn>copy Tumble.png HBO-ICT\YEAR1\QUARTILE1\OrganizationAndIT
1 file(s) copied.

C:\Saxionn>
```

Relevant screenshots **tree** command:

```
C:\Windows\System32\cmd.exe
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

C:\Saxionn>copy Wave.png YEAR1\QUARTILE1\IntroductionToProgramming
The system cannot find the path specified.
0 file(s) copied.

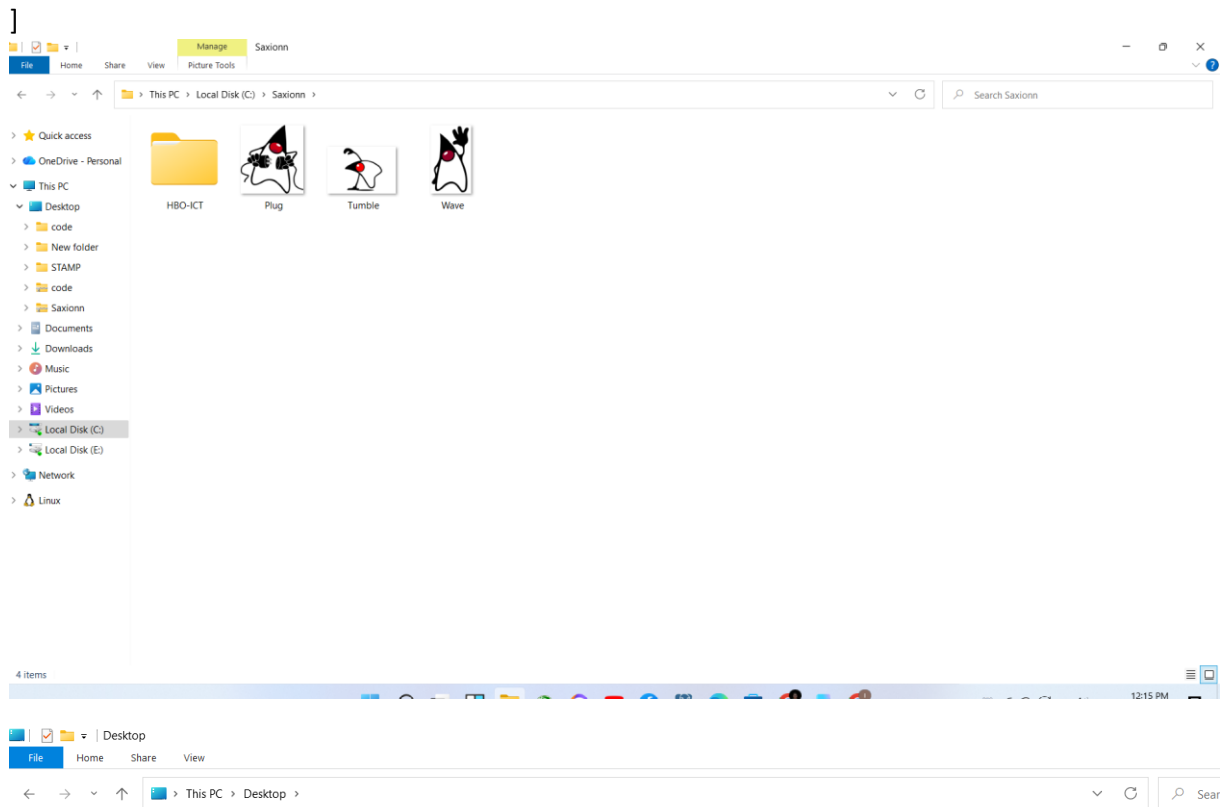
C:\Saxionn>copy Wave.png HBO-ICT\YEAR1\QUARTILE1\IntroductionToProgramming
1 file(s) copied.

C:\Saxionn>
C:\Saxionn>copy Plug.png HBO-ICT\YEAR1\QUARTILE1\IntroductionInfrastructure
1 file(s) copied.

C:\Saxionn>copy Tumble.png HBO-ICT\YEAR1\QUARTILE1\OrganizationAndIT
1 file(s) copied.

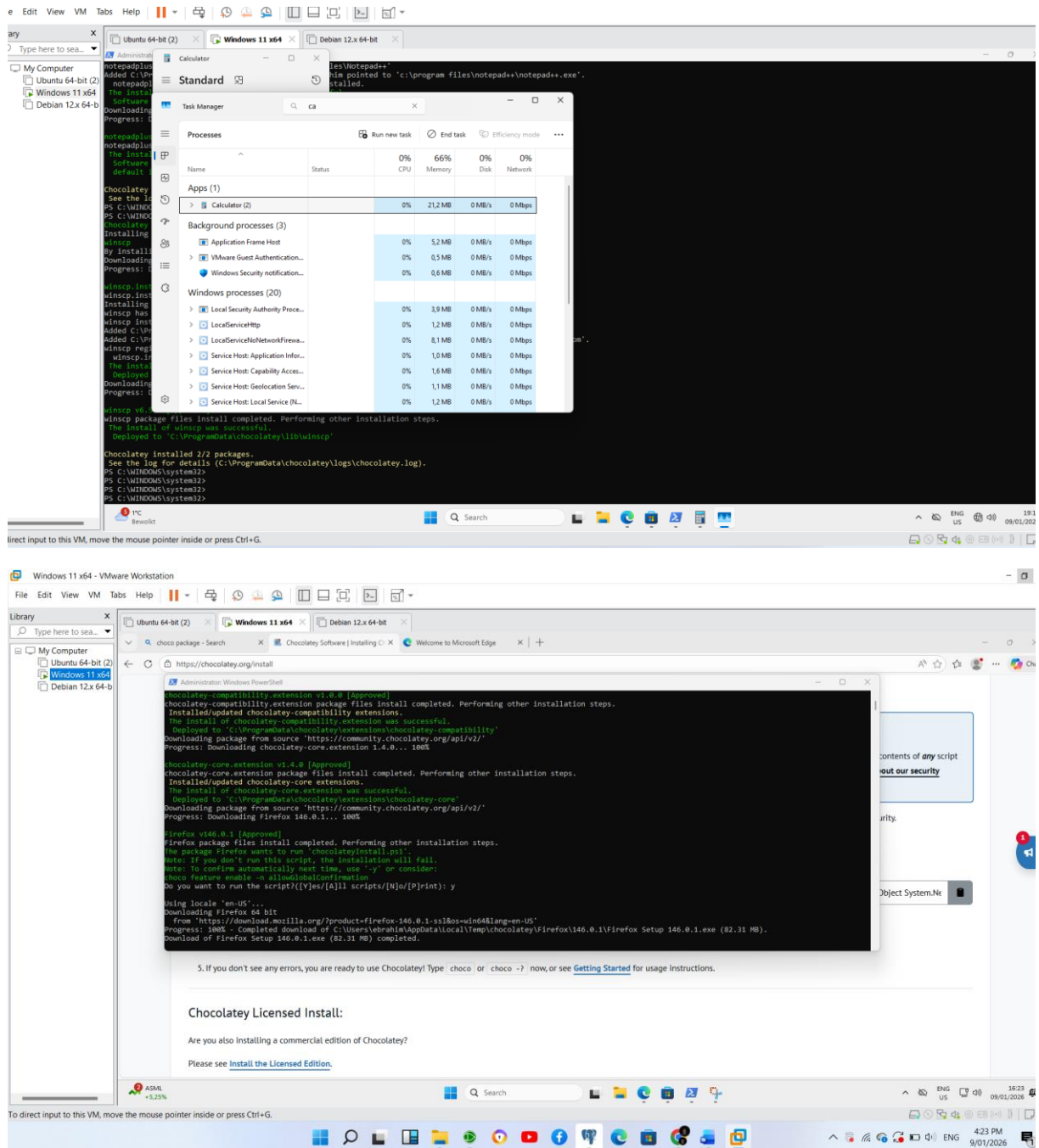
C:\Saxionn>tree
Folder PATH listing
Volume serial number is 2C67-AC72
C:.
|_ HBO-ICT
|   |_ YEAR1
|       |_ QUARTILE1
|           |_ IntroductionToProgramming
|           |_ ITInfrastructure
|           |_ Synergy
|           |_ QUARTILE2
|           |_ QUARTILE3
|           |_ QUARTILE4
|       |_ YEAR2
|           |_ QUARTILE1
|           |_ QUARTILE2
|           |_ Databases
|           |_ ITFundamentals
|           |_ ITGame
|           |_ QUARTILE3
|           |_ QUARTILE4
|       |_ YEAR3
|       |_ YEAR4
|
|_ C:\Saxionn>echo %username%
free bytes
C:\Saxionn>
```

Relevant screenshots in the file explorer of the folder c:\Saxion + created zip file.



Terminating Processes

Relevant Screenshots Task Manager Window:



Install Software

Relevant screenshots that the following software is installed with winget:

- WinSCP
- Notepad++
- 7zip

```

} Ubuntu 64-bit (2)
} Windows 11 x64
} Debian 12 x64-b

7zip.install has been installed.
7zip installed to 'C:\Program Files\7-Zip'
Added C:\ProgramData\chocolatey\bin\7z.exe shim pointed to 'c:\program files\7-zip\7z.exe'.
7zip.install can be automatically uninstalled.
The install of 7zip.install was successful.
Deployed to 'C:\Program Files\7-Zip'
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading 7zip 25.1.0... 100%

7zip v25.1.0 [Approved]
7zip package files install completed. Performing other installation steps.
The install of 7zip was successful.
Deployed to 'C:\ProgramData\chocolatey\lib\7zip'

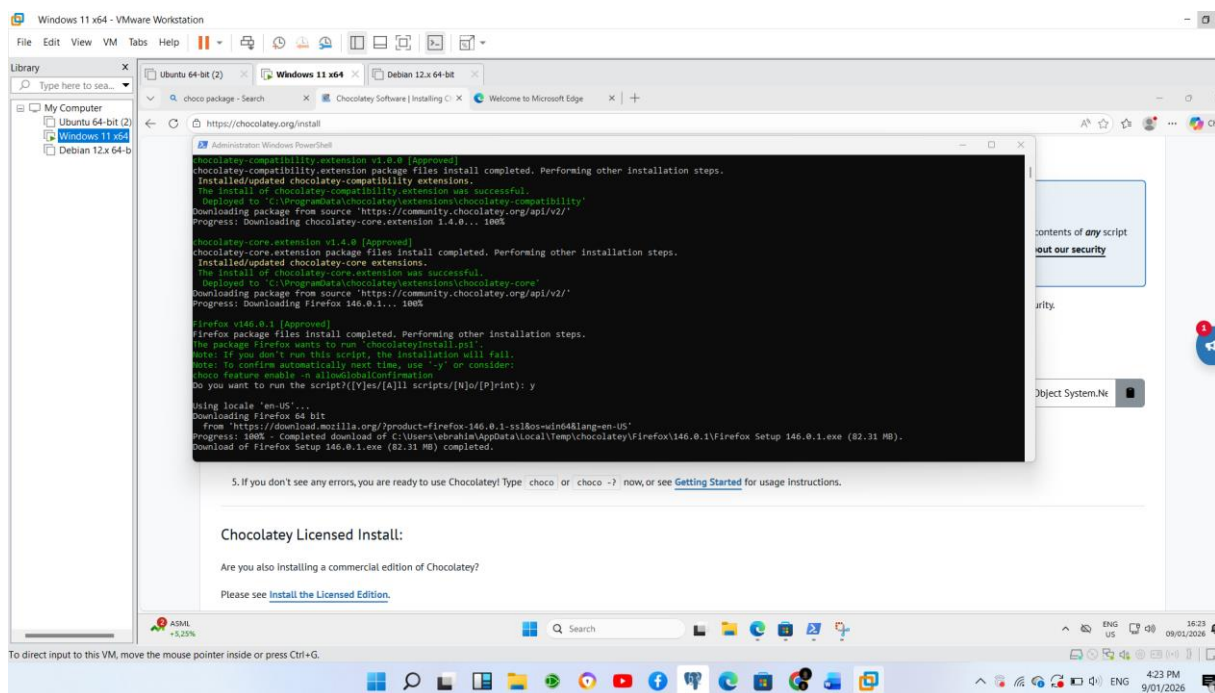
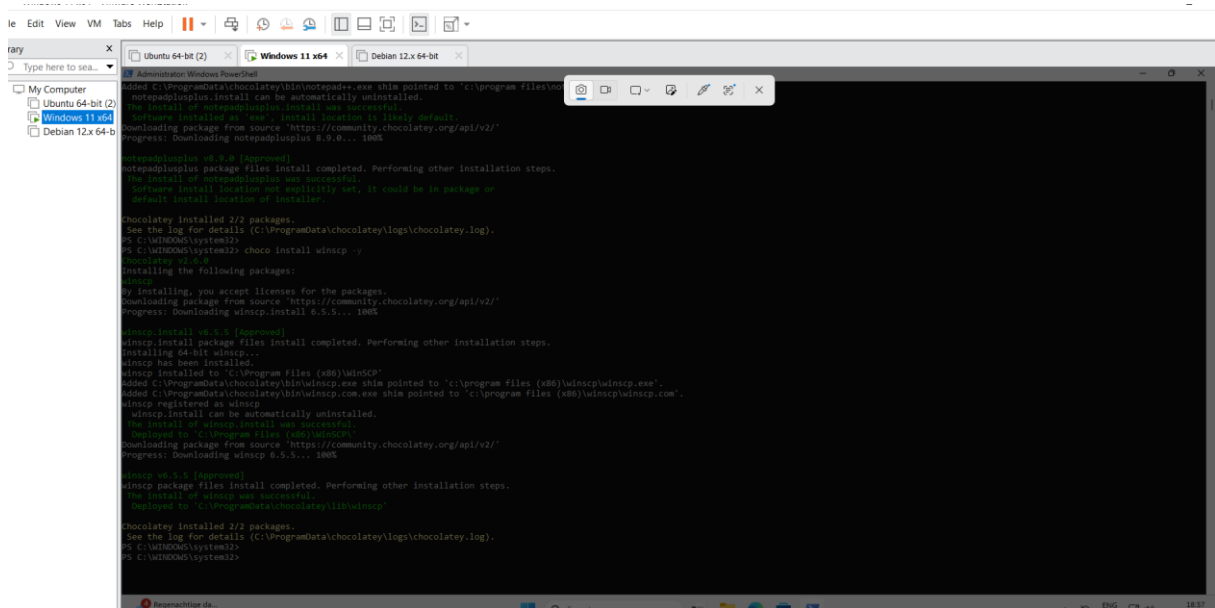
Chocolatey installed 2/2 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> choco install notepadplusplus -y
Chocolatey v2.6.0
Installing the following packages:
notepadplusplus
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus.install 8.9.0... 100%

notepadplusplus.install v8.9.0 [Approved]
notepadplusplus.install package files install completed. Performing other installation steps.
Installing 64-bit notepadplusplus.install...
notepadplusplus.install has been installed.
WARNING: No registry key found based on 'Notepad++\*'
notepadplusplus.install installed to 'C:\Program Files\Notepad++'
Added C:\ProgramData\chocolatey\bin\notepad++.exe shim pointed to 'c:\program files\notepad++\notepad++.exe'.
notepadplusplus.install can be automatically uninstalled.
The install of notepadplusplus.install was successful.
Software installed as 'exe', install location is likely default.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus 8.9.0... 100%

notepadplusplus v8.9.0 [Approved]
notepadplusplus package files install completed. Performing other installation steps.
The install of notepadplusplus was successful.
Software install location not explicitly set, it could be in package or
default install location of installer.

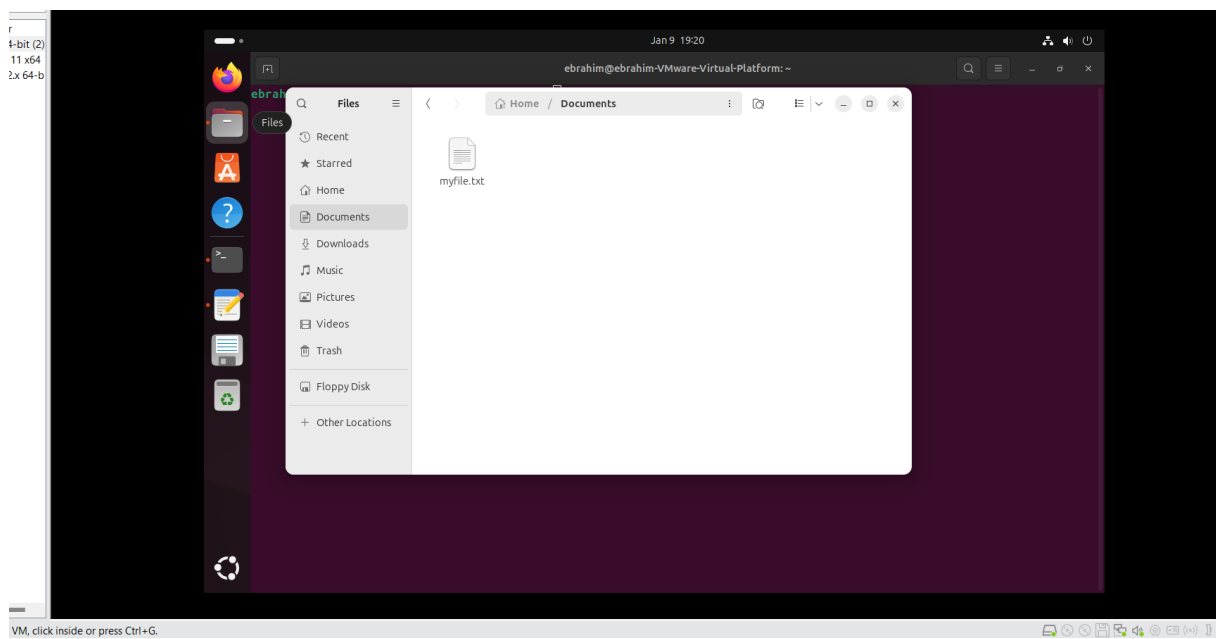
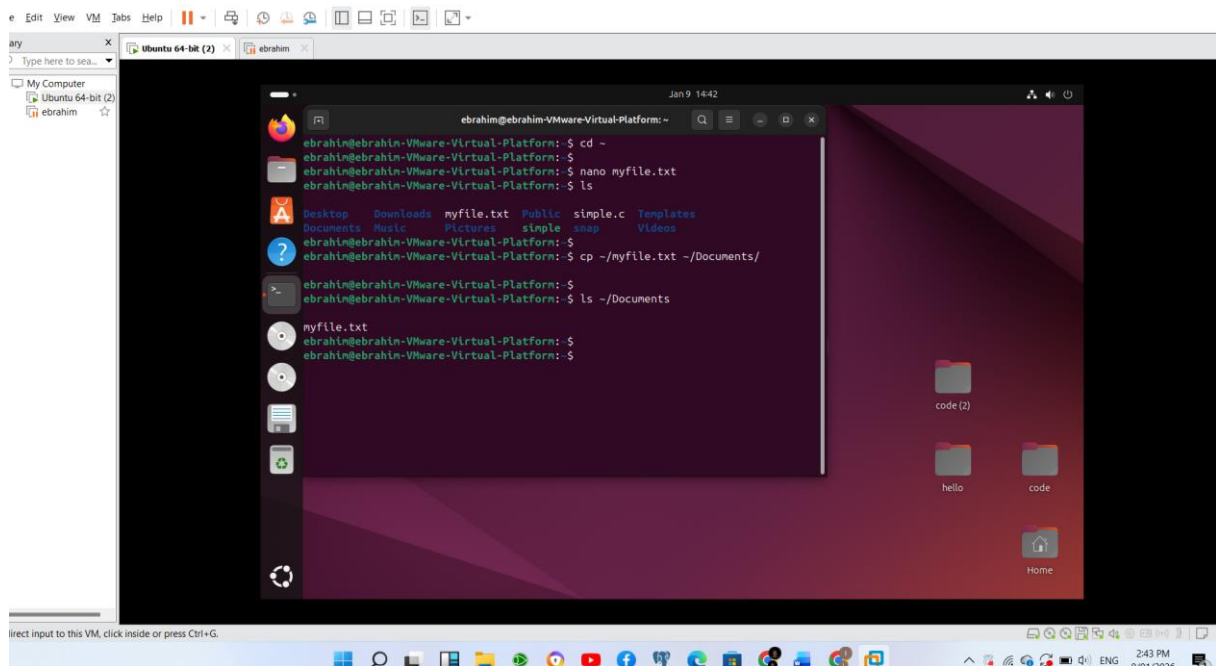
Chocolatey installed 2/2 packages.
See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32> choco install winscp -y
Chocolatey v2.6.0

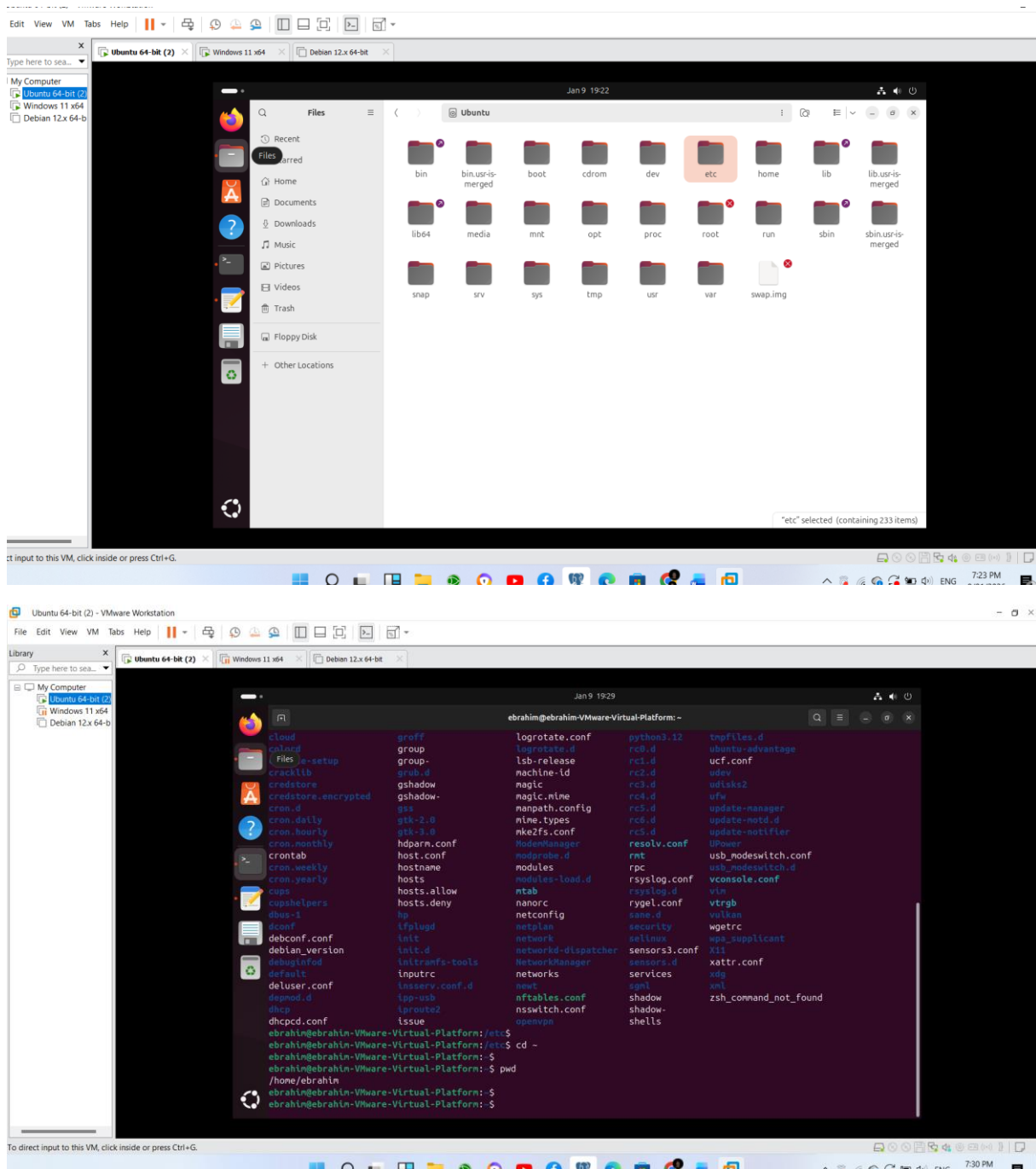
```



Assignment 5.4: Working with Linux

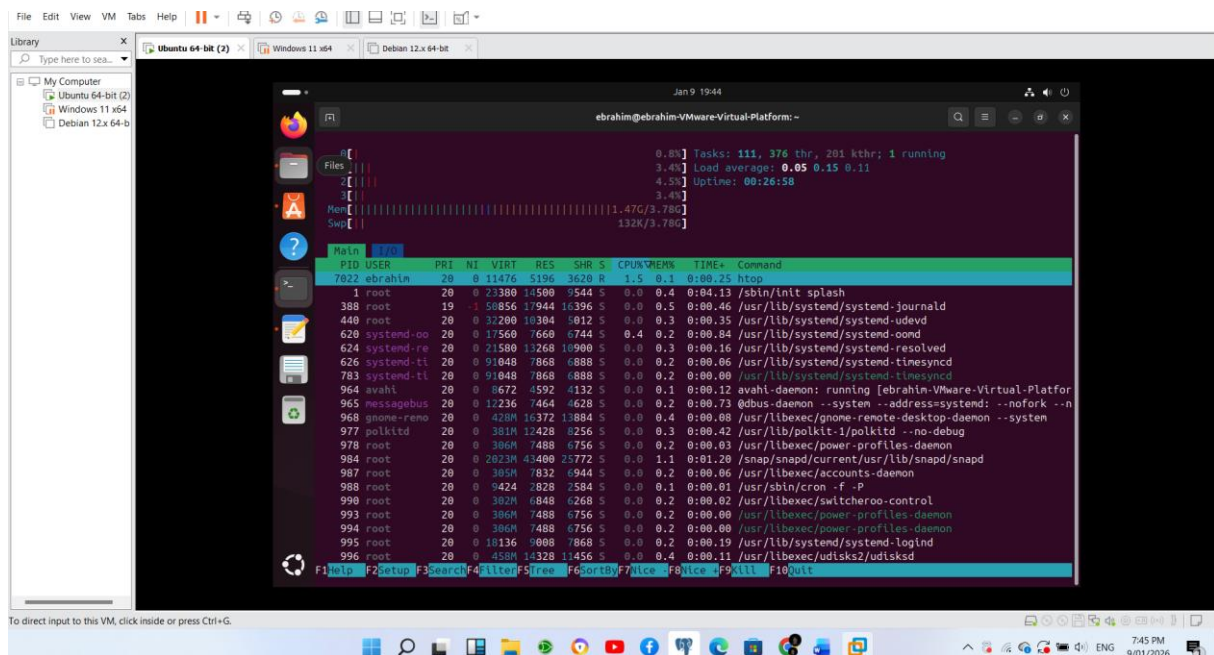
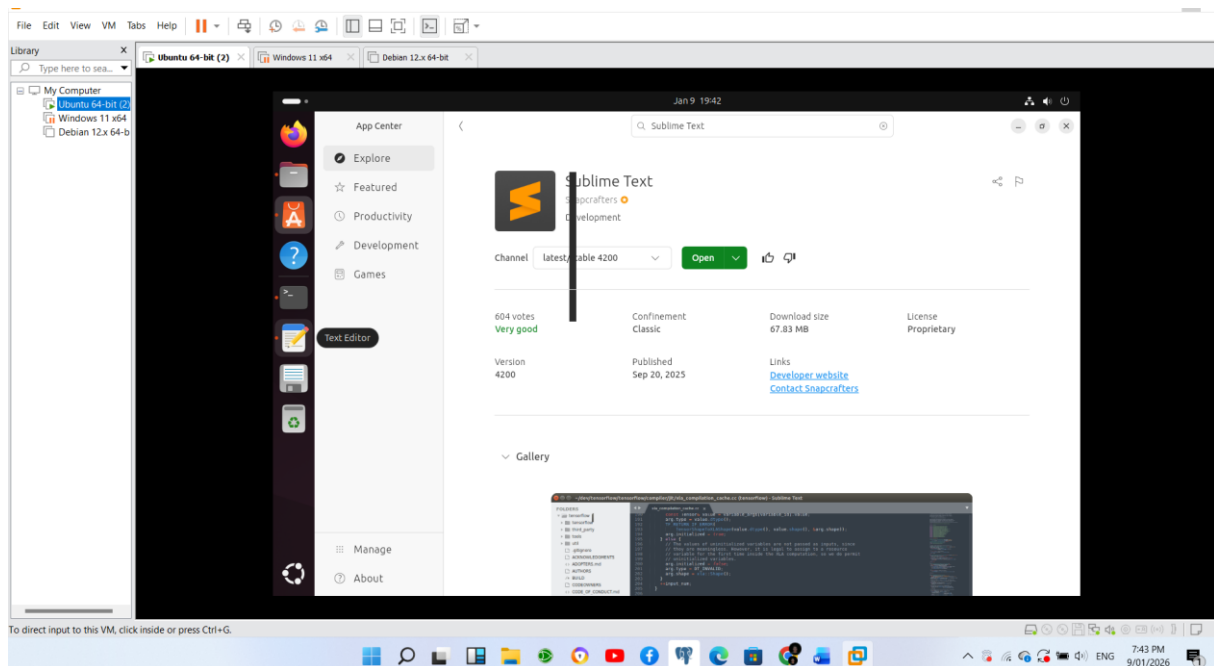
Relevant screenshots + motivation






```
ebrahim@ebrahim-Virtual-Platform: ~  
$ pwd  
/home/ebrahim  
$ cd -  
$  
$ tar -cvf myfile.tar myfile.txt  
myfile.txt  
$  
$ tar -czvf myfile.tar.gz myfile.txt  
myfile.txt  
$  
$ ls  
Downloads  myfile.tar  myfile.txt  Public  simple.c  Templates  
Music      myfile.tar.gz  Pictures  simple  snap     Videos  
$  
$ tar -cvf file.tar file.txt  
tar: file.txt: Cannot stat: No such file or directory  
tar: Exiting with failure status due to previous errors  
$  
$ tar -czvf myfile.tar.gz myfile.txt  
myfile.txt  
$  
$ tar -czvf myfile.tar.gz myfile.txt  
myfile.txt  
$  
$
```

```
Get:27 http://nl.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [387 kB]  
Get:28 http://nl.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]  
Get:29 http://nl.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.5 kB]  
Get:30 http://nl.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [30.3 kB]  
Get:31 http://nl.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [6,048 B]  
Get:32 http://nl.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]  
Get:33 http://nl.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [488 B]  
Get:34 http://nl.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]  
Get:35 http://nl.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,308 B]  
Get:36 http://nl.archive.ubuntu.com/ubuntu noble-backports/main Icons (48x48) [9,536 B]  
Get:37 http://nl.archive.ubuntu.com/ubuntu noble-backports/main Icons (64x64) [11.3 kB]  
Get:38 http://nl.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]  
Get:39 http://nl.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]  
Get:40 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [29.5 kB]  
Get:41 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [17.9 kB]  
Get:42 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]  
Get:43 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe Icons (48x48) [23.3 kB]  
Get:44 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe Icons (64x64) [32.3 kB]  
Get:45 http://nl.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,444 B]  
Get:46 http://nl.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]  
Fetched 13.7 MB in 4s (3,820 kB/s)  
  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
92 packages can be upgraded. Run 'apt list --upgradable' to see them.  
$
```



This assignment shows that I successfully practiced basic Linux skills. I created and copied files using both the file explorer and the terminal, navigated through the Linux file system, and learned the difference between Linux and Windows folder structures. I also compressed and extracted files using tar and gzip, monitored system processes with htop, and installed software using both the Software Center and the terminal. The screenshots prove that I completed each task correctly and understand the basic Linux commands and system usage.

Assignment 5.5: Users and permissions on Linux

Relevant screenshots + motivation

```
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello
ebrahim@ebrahim-VMware-Virtual-Platform: $ mkdir ~/hello
ebrahim@ebrahim-VMware-Virtual-Platform: $ cd ~/hello
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ nano hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ cat hello.sh
#!/bin/bash
echo Hello Ebrahim 577534!
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ nano hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ chmod +x hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ ls -l hello.sh
-rwxrwxr-x 1 ebrahim ebrahim 44 Jan  9 19:52 hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ ./hello.sh
Hello Ebrahim 577534!
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ chmod 700 hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$ ls -l hello.sh
-rwx----- 1 ebrahim ebrahim 44 Jan  9 19:52 hello.sh
ebrahim@ebrahim-VMware-Virtual-Platform: ~/hello$
```

I created a bash script in ~/hello and made it executable using chmod.
I executed the script using ./hello.sh.
Using numeric chmod (700), only the logged-in user has execute permissions.

Assignment 5.6: View the contents of files

Relevant screenshots + motivation

cat → Displays the full contents of a file.

wc → Counts lines, words, and characters in a file.

less → Opens a file for scrolling and reading page by page.

head → Shows the first lines of a file.

tail → Shows the last lines of a file.

grep → Searches for text inside a file.

```
ebrahim@ebrahim-Virtual-Platform:~$ cd ~
ebrahim@ebrahim-Virtual-Platform:~$ wget https://www.gutenberg.org/files/1661/1661-0.txt -O SherlockHolmes.txt
--2026-01-09 19:57:48-- https://www.gutenberg.org/files/1661/1661-0.txt
| App Center www.gutenberg.org (www.gutenberg.org)... 152.19.134.47, 2618:28:3898:0:bad:cafe:47
Connecting to www.gutenberg.org (www.gutenberg.org)[152.19.134.47]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 607504 (593K) [text/plain]
Saving to: 'SherlockHolmes.txt'

SherlockHolmes.txt      100%[=====] 593.27K  1.04MB/s   in 0.6s

2026-01-09 19:57:49 (1.04 MB/s) - 'SherlockHolmes.txt' saved [607504/607504]

ebrahim@ebrahim-Virtual-Platform:~$ ls
Desktop  Downloads  hello  myfile.tar  myfile.txt  Public  simple  snap  Videos
Documents  Files  Home  myfile.tar.gz  Pictures  SherlockHolmes.txt  simple.c  Templates

ebrahim@ebrahim-Virtual-Platform:~$ wc SherlockHolmes.txt
12306 107562 607504 SherlockHolmes.txt
ebrahim@ebrahim-Virtual-Platform:~$ grep -n kingdom SherlockHolmes.txt
498:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
ebrahim@ebrahim-Virtual-Platform:~$ sed -n '3490,3510p' SherlockHolmes.txt
already in the Bermuda Dockyard, so that there is really no tie between
them. I think that that bit of news has consoled young McCarthy for all
that he has suffered."
```

```
ebrahim@ebrahim-Virtual-Platform:~$ wc SherlockHolmes.txt
12306 107562 607504 SherlockHolmes.txt
ebrahim@ebrahim-Virtual-Platform:~$ grep -n kingdom SherlockHolmes.txt
498:"I tell you that I would give one of the provinces of my kingdom to
1124:And that was how a great scandal threatened to affect the kingdom of
ebrahim@ebrahim-Virtual-Platform:~$ sed -n '3490,3510p' SherlockHolmes.txt
already in the Bermuda Dockyard, so that there is really no tie between
them. I think that that bit of news has consoled young McCarthy for all
that he has suffered."

"But if he is innocent, who has done it?"

"Ah! who? I would call your attention very particularly to two points.
One is that the murdered man had an appointment with someone at the
pool, and that the someone could not have been his son, for his son was
away, and he did not know when he would return. The second is that the
murdered man was heard to cry 'Cooee!' before he knew that his son had
returned. Those are the crucial points upon which the case depends. And
now let us talk about George Meredith, if you please, and we shall
leave all minor matters until to-morrow."

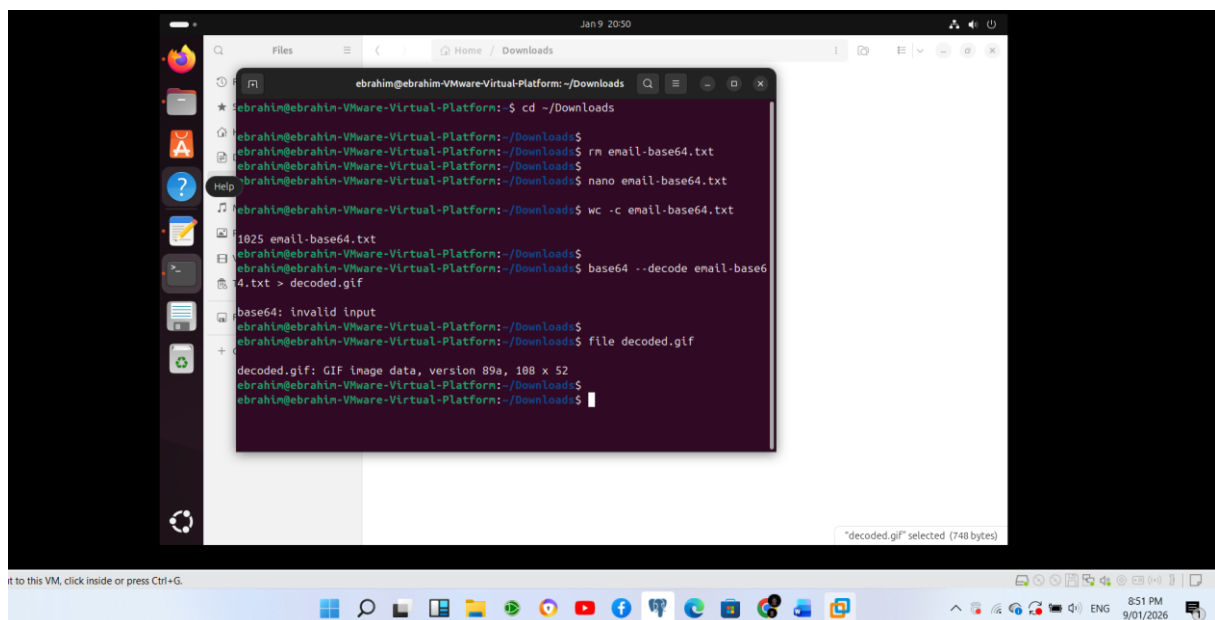
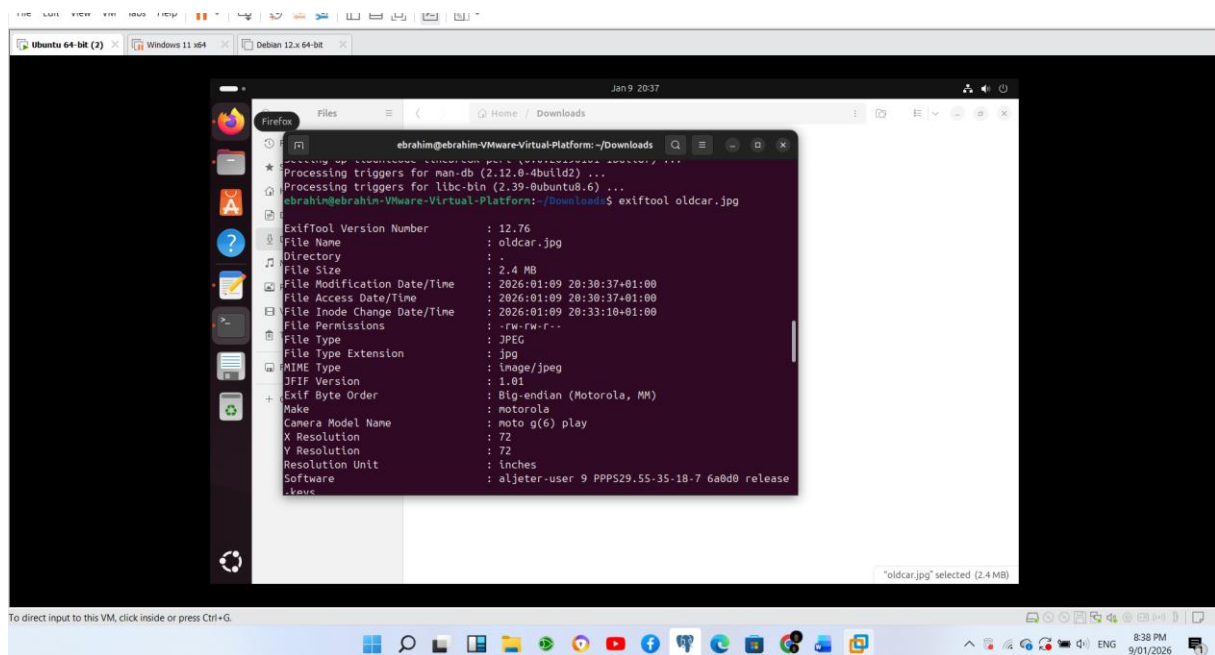
There was no rain, as Holmes had foretold, and the morning broke bright
and cloudless. At nine o'clock Lestrade called for us with the
carriage, and we set off for Hatherley Farm and the Boscombe Pool.

"There is serious news this morning," Lestrade observed. "It is said
that Mr. Turner, of the Hall, is so ill that his life is despaired of."
```

I used the wc command to count the number of lines, words, and characters in the file.
I used grep -n to find the word "kingdom" and display the line numbers.
Using text viewing commands, I displayed the lines around the word "kingdom".

Assignment 5.7: Digital forensics

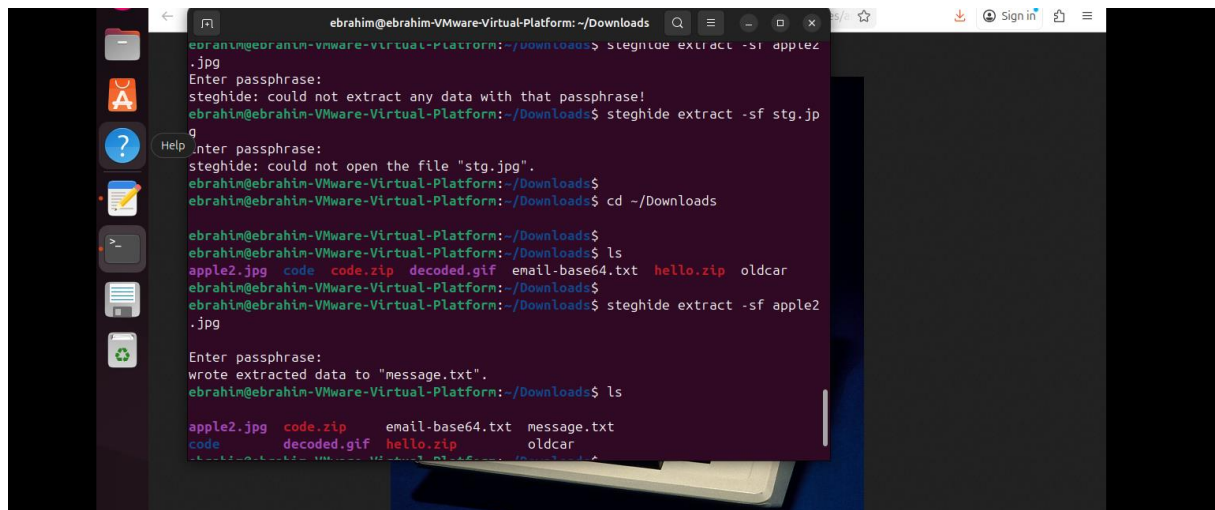
Relevant screenshots + motivation



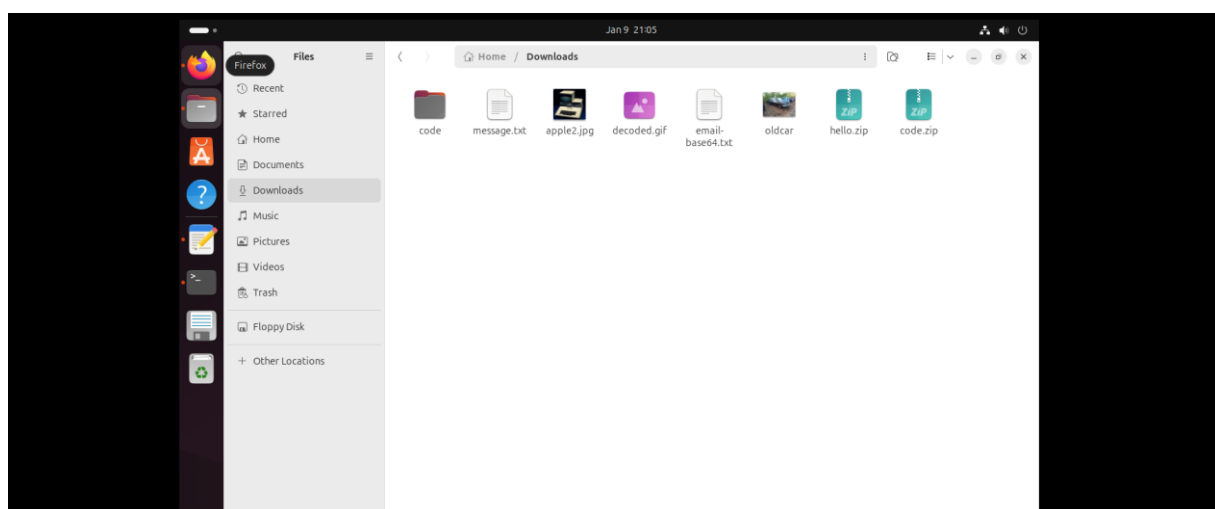
In this assignment, I practiced basic digital forensics techniques on Linux. I analyzed a JPEG file using the EXIF tool to identify device information and GPS metadata, and verified how Ubuntu recognizes file types even when the file extension is removed. I also worked with BASE64 encoding by decoding an intercepted text file and saving it as a binary GIF file using terminal commands and output redirection. This assignment helped me understand how metadata, file signatures, and binary encoding are used in digital investigations.

Assignment 5.8: Steganography

Relevant screenshots + motivation



```
ebrahim@ebrahim-VMware-Virtual-Platform: ~/Downloads
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ steghide extract -sf apple2.jpg
Enter passphrase:
steghide: could not extract any data with that passphrase!
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ steghide extract -sf stg.jpg
q
Enter passphrase:
steghide: could not open the file "stg.jpg".
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ cd ~/Downloads
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ ls
apple2.jpg  code  code.zip  decoded.gif  email-base64.txt  hello.zip  oldcar
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ steghide extract -sf apple2.jpg
Enter passphrase:
wrote extracted data to "message.txt".
ebrahim@ebrahim-VMware-Virtual-Platform:~/Downloads$ ls
apple2.jpg  code  code.zip  email-base64.txt  message.txt
code  decoded.gif  hello.zip  oldcar
```

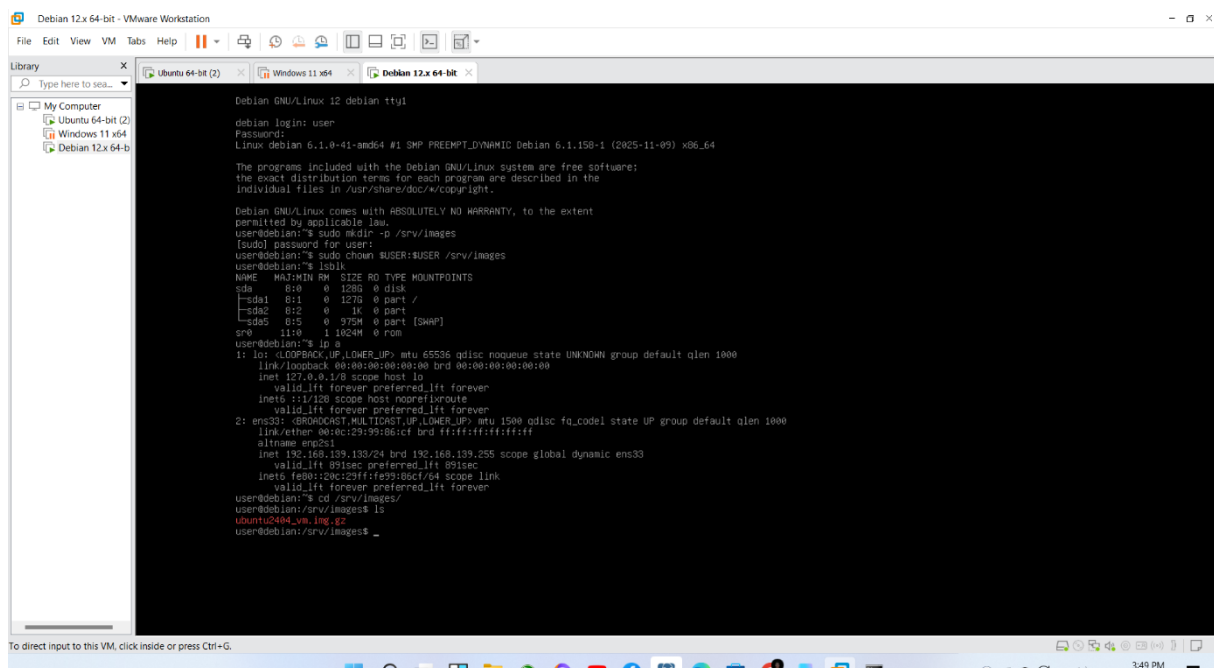
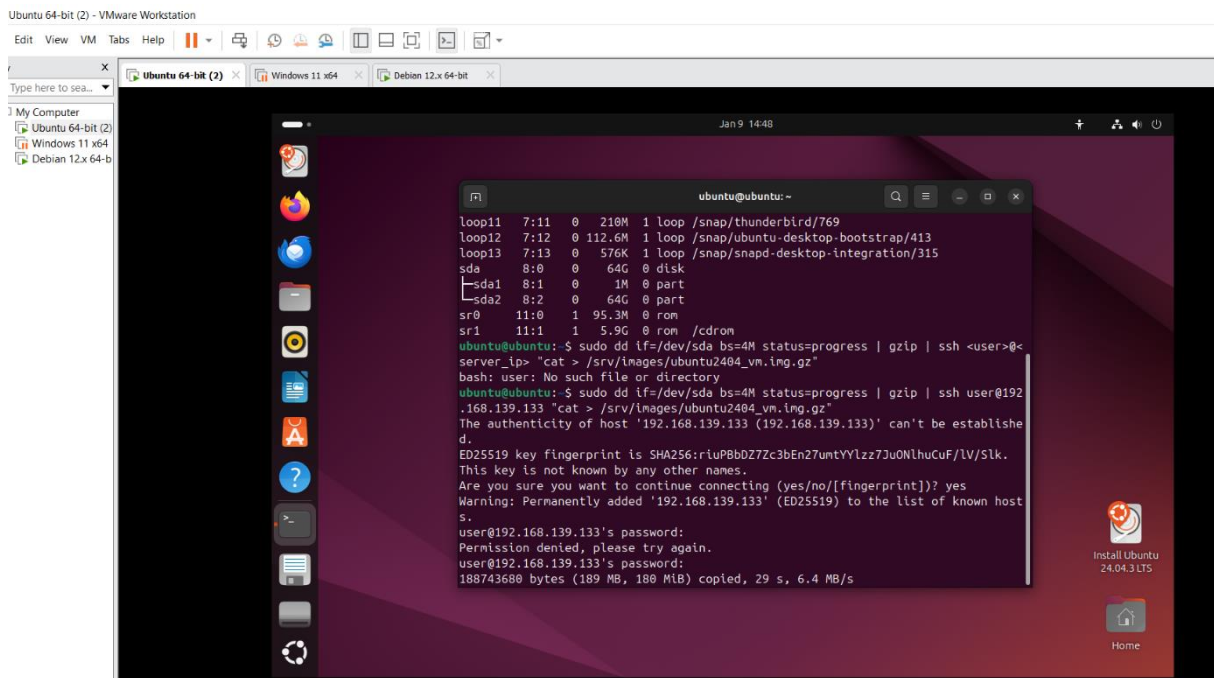


In this assignment, I learned how steganography works by hiding and extracting data inside an image file. Using the steghide command-line tool, I extracted a hidden text file from an image by applying the correct password. This showed how information can be stored in the least significant bits of a file without visibly changing the image. The assignment helped me understand how hidden data can be detected and recovered in digital forensics.

Assignment 5.9: Capture disk images

Make relevant screenshots + motivation:

- Proof that the Debian 13 server stored a back-up image of the Ubuntu 24.04 Desktop VM.
- Proof that you can restore the back-up image into an empty VM.



In this assignment, I learned how to capture and restore a complete virtual machine disk using Linux command-line tools. I configured a Debian server as an image server with SSH, created a storage directory for disk images, and verified network connectivity. From a live Ubuntu environment, I captured the full disk using `dd`, compressed it with `gzip`, and securely transferred it to the Debian server. I then restored the image onto a new empty virtual machine and successfully booted the recovered system. This assignment improved my understanding of system backups, disk imaging, data recovery, and enterprise deployment workflows.

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